

Multimodal Activity Detection for Natural Interaction with Virtual Human

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ABSTRACT

Natural face-to-face human-robot conversation is one of the most important features for virtual human in virtual reality and metaverse. However, the unintended wake-up of robot is often activated with only Voice Activity Detection (VAD). To address this issue, we propose a Multimodal Activity Detection (MAD) scheme, which considers not only voice but also gaze, lip-movement and talking content to decide whether the person is activating the robot. A dataset for large screen-based virtual human conversation is collected from various challenging cases. The experimental results show that the proposed MAD greatly outperforms VAD-only method.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction techniques

1 INTRODUCTION

It is natural to pay attention to the counterpart's voice, facial expression and semantic meaning when people are talking in a face-to-face manner. Similarly, when people are talking with a robot, such as a digital human in a big screen or a humanoid robot with similar height, the multimodal information should also be considered to reach natural communication [1]. Unfortunately, this has not been totally achieved for human-robot conversations, and unintended wake-up of robot is often activated as shown in Fig. 1. For example, a robot at a noisy exhibition usually gives unexpected reactions when it is having a conversation with a visitor, as it may collect the sound of other human conversations and mistake it as the words speaking to itself. When a digital human talking with a customer, it may give wrong reactions when taking the topics that other people talk as those speaking to it.

In current human-robot conversations, Voice Activity Detection (VAD) is often adopted to judge if their exist the sound of talking and activate the robot's reactions. This strategy may work for the chatbot with one person in short-distance interaction, while for face-to-face conversations it usually fails. When the speech comes from the environment or other people instead of the one who is talking to the robot, or when it is the voice that the user talking to others instead of the robot, the robot may give wrong reactions. Therefore, a new Activity Detection scheme is needed in human-robot conversations to solve these problems.

In this paper, we propose a Multimodal Activity Detection (MAD) scheme which includes multiple factors like voice, pose,

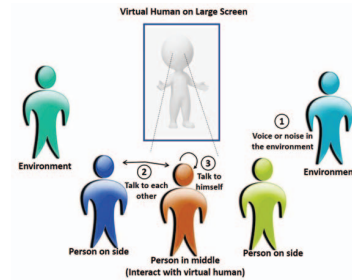


Figure 1: Unintended wake-up of virtual human in face-to-face human-machine conversation.

lip-movement, the content of talking and so on. It simulates the wake-up policy in face-to-face conversations, and activates the robot through fusing the multimodal information and making decisions. Comparing to the single-modality VAD scheme, it is able to significantly reduce the wrong activation of robots.

2 METHOD

The MAD is implemented in time segments, in which the multimodal information will be processed independently first and then combined to make the decision, as shown in Fig. 2.

2.1 Voice Activity Detection (VAD)

In this modal, the robot conversation is activated when the person's voice is detected. Generally, VAD methods consist of two modules: feature extraction and activity decision. For feature extraction, the fluctuation of energy or zero crossing rate is often considered in temporal domain of voice, while the spectrum or cepstrum is considered in frequency domain of voice. For activity decision, three kinds of method are often used, i.e., threshold based method, statistical model-based method, and deep learning based method. Here, the method [4] is employed that computes the cepstrum MFCCs of voice segments and feeds them into a RNN based classifier. The classifier tells the existence of voice activity segment by segment.

2.2 Pose Activity Detection (PAD)

In this modal, the robot conversation is activated when the person is gazing at the robot. The Kinect sensor-based method [7] has already proved to be robust to environment changes and is thus employed here. Specifically, the 3D skeleton of the person is acquired from the Kinect sensor and the 3D head position of the person is obtained directly from the skeleton, then the corresponding head area is cropped from the RGB image and is fed into a label distribution learning based CNN network [5] to robustly and accurately estimate the head pose. Combining the head position and head pose of the person with the robot size, we can easily determine whether the person is focusing his eyes on the robot.

2.3 Lip Activity Detection (LAD)

In this modal, the robot conversation is activated when talking-related lip movements are detected. Here first detect the face and

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the 68 landmarks by using InsightFace [3]. The we select the 9 landmarks around lip area and calculate the average pixel separations between part pairs and store away this distance value into the lip separation sequence. They are then fed into the RNN [6]MJ99 network for consecutive context extraction and output the classification estimation, which indicating the respective probability of whether the speaker was speaking or was silent during the preceding frames.

2.4 Context Activity Detection (CAD)

In this modal, the robot conversation is activated when the person is asking the robot some question. To do it, a text based wake-up model is trained to tell whether the query text converted from voice by Automatic Speech Recognition (ASR) is a question sentence needing the robot's response. This model is composed of a pretrained BERT model [2] for feature extraction followed with a full connection layer for classification. The full connection layer is trained with the two-class dataset consisting of question sentences and non-question sentences.

2.5 Multimodal Activity Detection (MAD)

The above four introduced activity detection modules are jointly used to judge if a robot should response or not, as shown in 2. For the fusion of these activities, there are two strategies: the decision-level fusion and the feature-level fusion. In the decision-level fusion, the results of different modalities are fused in a time segment.

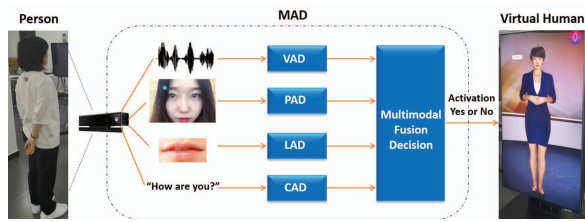


Figure 2: The proposed MAD scheme composed of VAD, PAD, LAD and CAD.

3 EXPERIMENTS

Prototype and Dataset We implement the prototype of large-screen based virtual human conversation system as shown in Fig. 3(a). It is composed of a large-screen to display person-sized virtual human, the Kinect on top of screen to capture voice, rgb and depth information in real time, and GPU-equipped computer for MAD and interaction. Based on the prototype, we collect 100 videos from various interaction scenarios as the dataset. For each video, the VAD method is applied to tell the potential activity points in voice automatically, based on which, the ground-truth points are confirmed manually to avoid some wrong points. Totally, there are about 600 effective activity points. Additionally, for each activity point, the corresponding activity status in other modalities are labelled, including PAD, LAD and CAD.

Specially, to show the advantage of MAD, some challenging interaction cases (as shown in Fig. 3(b)) are considered when constructing the dataset. Here, the person in middle denotes the person in focus of virtual human's viewpoint. To make the comparison of various modality's performances easier, we set that only the person in middle can activate the virtual human.

Experimental Results Taking two thirds of dataset for training or finetuning the deep learning models in VAD, PAD, LAD and CAD, several decision-level fusion methods are tested on the remaining one third of dataset. The comparative results are given in Table 1. As can be seen the more the number of modality to fuse, the better performance the fusion method obtains. And, the MAD fusing all

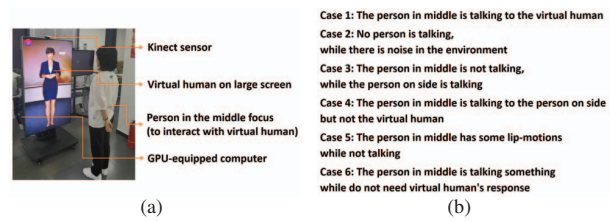


Figure 3: The implemented prototype of large-screen based virtual human conversation system (a) and the interaction cases set in the dataset collection (b).

the four modalities together gets the best accuracy 91% that is much better than others. The reason is that based on VAD's results, some false positive happening in Case 4 is removed by PAD, some false positive happening in Case 3 is removed by LAD, and some false positive happening in Case 6 is removed by CAD.

Fused Modalities	Accuracy
VAD only	68%
VAD+PAD	75%
VAD+PAD+LAD	84%
VAD+PAD+LAD+CAD	91%

Table 1: Comparison of different number of modality fusion in MAD.

4 CONCLUSION

This paper presents a Multimodal Activity Detection scheme to solve the mis-wakeup problem in face-to-face human-robot conversation. A dataset with various challenging conversation cases is constructed with an implemented prototype. Experimental results show that multimodal fusion methods outperform such single modal method as VAD. The dataset will be released for public research.

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